

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 20

AIR POLLUTION - 7

ASBESTOS

- Asbestos refers to a set of six naturally occurring silicate fibrous minerals -- chrysotile, crocidolite, amosite, anthophyllite, tremolite, and actinolite.
- Asbestos is flexible and resistant to heat and electricity. So, it is widely used to make construction materials, automotive parts etc.
- The fibers that form asbestos separate very easily into tiny pieces when they're handled or damaged. They're too small to see and easy to breathe in.
- Prolonged inhalation of asbestos fibers can cause serious and fatal illnesses including lung cancer, mesothelioma, and asbestosis.



ASBESTOS

- The Mesothelium is the layer of tissues that surrounds the organs of the chest (Heart and Lungs), abdominal cavity and pelvis. Its function is to protect the organs and to aid in breathing.
- Mesothelioma is a rare type of cancer that occurs in the mesothelium. Mesothelioma is caused by the inhalation of asbestos particles. In most cases, mesothelioma symptoms will not appear in an individual exposed to asbestos until many years after the exposure has occurred.
- Asbestosis is long term inflammation and scarring of the lungs due to asbestos.
- Symptoms may include
 - ✓ shortness of breath
 - ✓ cough and
 - ✓ chest pain.
- Complications may include
 - ✓ lung cancer and
 - ✓ mesothelioma.



RADON

- Radon is a powerful radioactive gas that is naturally emitted from soil when radioactive elements like uranium and thorium go through decay.
- Studies have shown a clear link between breathing high concentrations of radon and occurrence of lung cancer.
- Radon is a contaminant that affects indoor air quality worldwide.
- According to the United States Environmental Protection Agency (EPA), radon is the second most frequent cause of lung cancer, after cigarette smoking, causing 21,000 lung cancer deaths per year in the United States.
- Radon is heavier than air. It is found in low lying areas like basements and cellars and in poorly ventilated places.
- Due to modern houses having poor ventilation, it is confined inside the house and causes lung cancers.



SMOG

- Smog is smoke + fog (smoky fog). It is caused by the burning of large amounts of coal, vehicular emissions and industrial emissions.
- Smog contains
 - ✓ soot particles
 - ✓ smoke
 - ✓ sulphur dioxide
 - ✓ nitrogen oxides (NO_x) etc.
- Two distinct types of smog are recognized:
 - ✓ sulfurous smog and
 - ✓ photochemical smog.



SMOG

SULFUROUS SMOG

- Sulfurous smog is also called 'London smog' as it was detected in London for the first time.
- Sulfurous smog results from a high concentration of Sulfur Di Oxide (SO_2) in the air and is caused by the use of sulfur-bearing fossil fuels like diesel and particularly coal.
- Coal was the main source of power in industries around London during nineteenth century.
- This type of smog is aggravated by dampness and a high concentration of suspended particulate matter in the air.

PHOTOCHEMICAL SMOG

- Photochemical smog is also known as 'Los Angeles smog'.
- Photochemical smog occurs mostly in urban areas that have large numbers of automobiles as Nitrogen Oxides are emitted by them.



FORMATION OF PHOTO CHEMICAL SMOG

- When pollutants such as nitrogen oxides and volatile organic compounds react together in the presence of sunlight. Ground Level Ozone and peroxyacyl nitrate (PAN) are formed.
- Photochemical is a combination of Nitrogen Oxides, Ground level ozone, peroxyacyl nitrates (PAN) and particulate matter.
- For these reactions to require bright and strong solar radiation is required, so generally photochemical smog is formed in summers.
- So, it is known as summer smog.
- It is also called Haze; it causes reddish brown discoloration of the sky.

Haze is a condition in which particulate matter and aerosols obscure the visibility of the sky.

- Peroxyacyl nitrate (PAN) is a powerful respiratory and eye irritant present in photochemical smog.
- At higher concentrations they cause extensive damage to human health and vegetation. PAN also causes skin cancer.



SMOG

HEALTH EFFECTS

- Smog is highly toxic to humans and can cause severe health problems. Smog causes inflammation of respiratory tract, decreases the lungs' working capacity, causes shortness of breath, pain while inhaling and cough.
- It can cause eye and nose irritation and it damages the mucous membranes which are the protective membranes of the nose and throat and interferes with the body's ability to fight infection, increasing susceptibility to illness.
- Smog is a combination of airborne particulate matter, like soot, and toxic gases including ground level ozone (O_3), carbon monoxide (CO), sulfur dioxide (SO_2), which are carcinogens.

OTHER EFFECTS

- Smog reduces visibility drastically.
- Smog prevents pollutants from escaping out.



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